

Online Movie Ticket Booking Web Application

Product Backlog, Acceptance Criteria, and Non-Functional Requirements

Web application

Primary actors	Customer, Admin, Cinema Staff
Payment model	External payment provider handles payment processing; the application manages booking/payment state.
Seat hold	10-minute temporary hold; successful payment changes seats to BOOKED.
Cancellation	Customer cannot cancel a successfully purchased ticket; showtimes with confirmed bookings cannot be removed.

1. Domain Glossary

Term	Definition
Location	A geographical area in Vietnam used for browsing cinemas, such as a district or ward.
Cinema	A specific movie theater located inside a Location.
Hall	A screening room belonging to a Cinema.
Seat Layout	A predefined layout template describing the physical seats inside a Hall.
Seat	A physical position in a Hall, such as E5.
Movie	A film available for screening, including details such as title, description, duration, poster, genre, and age rating.
Showtime	A scheduled screening of a Movie in a specific Cinema and Hall at a specific date and time.
Seat Hold	A temporary reservation of selected seats for up to 10 minutes while a customer completes payment.
Booking	The customer transaction for selected seats; it becomes confirmed after successful payment.
Payment Attempt	One attempt to pay for a Booking. A Booking may have multiple attempts, but only one active attempt at a time.
Ticket	The confirmed proof of purchase containing booking information and a QR code.
Customer	A registered user who browses movies and purchases tickets.
Admin	A role that manages movie, venue, and showtime information.
Cinema Staff	An authorized user who validates tickets at the cinema entrance.

Domain rule: A movie is considered available at a Location because it has at least one configured showtime at a Cinema in that Location; the system does not maintain a separate movie-location availability flag.

2. Product Backlog

Priority notation: M = Must have for MVP; S = Should have in Release 2. Estimates are relative story points.

ID	Epic	Backlog Item	Release	Dependencies	Est.
ID-01	Identity	Customer registration	M / MVP	—	3
ID-02	Identity	User login and logout	M / MVP	ID-01	3
ID-03	Identity	Password reset	S / R2	ID-01	3
ID-04	Identity	Role-based access for Customer, Admin, and Cinema Staff	M / MVP	ID-02	5
ID-05	Identity	Customer updates personal profile information	MVP	ID-02	5
CAT-01	Catalog	Admin creates and edits movie information	M / MVP	ID-04	5
CAT-02	Catalog	Customer browses, searches, filters, and views movie details	M / MVP	CAT-01	5
CAT-03	Catalog	Admin archives a movie when it has no future showtimes	S / R2	CAT-01, SCH-01	3
VEN-01	Venue	Admin manages configured locations, cinemas, and halls	M / MVP	ID-04	5
VEN-02	Venue	Admin assigns a predefined seat layout to a hall	M / MVP	VEN-01	3
SCH-01	Scheduling	Admin creates and manages showtimes	M / MVP	CAT-01, VEN-01, VEN-02	5
BKG-01	Booking	Customer views showtimes by location and cinema	M / MVP	SCH-01	3
BKG-02	Booking	Customer views seat layout and seat status	M / MVP	SCH-01, VEN-02	3
BKG-03	Booking	Customer holds selected seats for 10 minutes	M / MVP	BKG-02, ID-02	5
BKG-04	Booking	System resolves concurrent seat reservations	M / MVP	BKG-03	5
PAY-01	Payment	Customer creates and retries payment attempts for a booking	M / MVP	BKG-03	5
PAY-02	Payment	System processes payment outcomes and reconciles uncertain results	M / MVP	PAY-01	8
TCK-01	Tickets	System generates and displays a confirmed ticket with QR code	M / MVP	PAY-02	5

ID	Epic	Backlog Item	Release	Dependencies	Est.
TCK-02	Tickets	Customer views upcoming tickets and recent booking history	M / MVP	TCK-01	3
TCK-03	Ticket Validation	Cinema Staff validates customer tickets using QR codes	M / MVP	TCK-01, ID-04	5

3. Acceptance Criteria

Each backlog item below is expressed in Given/When/Then form so the requirement can be implemented and tested.

ID-01 Customer registration

- **Given** the customer is not registered, **when** they submit valid required information, **then** the system creates a customer account.
- **Given** the submitted email already belongs to an account, **when** the customer submits the registration form, **then** the system rejects registration and explains that the account already exists.
- **Given** required information is missing or invalid, **when** the customer submits the form, **then** the system displays validation errors and does not create the account.
- **Given** a customer registers with a new email address, **when** registration succeeds, **then** the email is associated with the created customer account and must satisfy the project's email-verification policy.

ID-02 User login and logout

- **Given** a registered user enters valid credentials, **when** they submit the login form, **then** the system authenticates the user and creates an authenticated session.
- **Given** invalid credentials are entered, **when** the user attempts to log in, **then** the system rejects authentication.
- **Given** an authenticated user selects Logout, **when** the logout request succeeds, **then** the session is terminated and protected functions require authentication again.
- **Given** an Admin or Cinema Staff seeded account enters valid credentials, **when** they log in, **then** the system authenticates the account using its assigned role.
- **Given** a customer has successfully changed and verified their email address, **when** they log in using the new email and correct password, **then** the system authenticates the same customer account.

ID-03 Password reset

- **Given** a registered customer requests a password reset, **when** the request is accepted, **then** the system starts the password-reset process.
- **Given** the customer provides valid reset information and a valid reset token, **when** they submit a new password, **then** the system updates the password.
- **Given** the reset token is invalid or expired, **when** the customer attempts to use it, **then** the system rejects the reset.
- **Given** a customer has changed and verified their email address, **when** they request a password reset, **then** the reset process uses the current verified email.

ID-04 Role-based access

- **Given** a Customer is authenticated, **when** they attempt to access an Admin or Staff function, **then** the system denies the request.
- **Given** an Admin is authenticated, **when** they access movie, venue, or showtime management functions, **then** the system permits those operations.
- **Given** Cinema Staff is authenticated, **when** they access ticket-validation functions, **then** the system permits those operations.
- **Given** an unauthenticated user accesses a protected function, **when** the request reaches the system, **then** access is denied.

ID-05 Update Customer Profile

- **Given** an authenticated customer submits valid editable profile information, **when** the update is submitted, **then** the system updates the profile.
- **Given** a customer enters an email already associated with another account, **when** the update is submitted, **then** the system rejects the email change.
- **Given** a customer enters a new valid email address, **when** the change is submitted, **then** the system sends a verification request to the new email and does not make it the active email until verification succeeds.
- **Given** the new email has been successfully verified, **when** the verification is completed, **then** the new email becomes the customer's active email and the customer account remains the same account.
- **Given** a customer has existing bookings or tickets, **when** the customer's email is changed, **then** those bookings and tickets remain associated with the same customer account.

CAT-01 Manage movie information

- **Given** an Admin is authenticated, **when** valid movie information is submitted, **then** the system creates the movie.
- **Given** an existing active movie, **when** the Admin submits valid edited information, **then** the system updates the movie.
- **Given** required movie information is invalid or missing, **when** the Admin submits the form, **then** the system rejects the operation and displays validation errors.
- **Given** a movie has future showtimes, **when** an edit would violate an existing scheduling constraint, **then** the system rejects the conflicting change.

CAT-02 Browse, search, filter, and view movie details

- **Given** active movies exist, **when** a Customer opens the catalog, **then** available active movies are displayed.
- **Given** a Customer selects a movie, **when** its detail page opens, **then** the system displays its configured movie details.
- **Given** multiple active movies exist, **when** the Customer searches or filters the catalog, **then** only matching movies are displayed.
- **Given** a movie is archived, **when** the Customer browses the active catalog, **then** that movie is not displayed as an available movie.

CAT-03 Archive a movie

- **Given** a movie has no future showtimes, **when** the Admin archives it, **then** its status changes to Archived.
- **Given** a movie has at least one future showtime, **when** the Admin attempts to archive it, **then** the operation is rejected.
- **Given** a movie is Archived, **when** the Admin attempts to create a new showtime for it, **then** the operation is rejected.
- **Given** an archived movie has historical tickets, **when** a Customer views those tickets, **then** the movie information remains available for historical reference.

VEN-01 Manage locations, cinemas, and halls

- **Given** the system contains configured Vietnamese Locations, Cinemas, and Halls, **when** the Admin opens venue management, **then** those entities are available for configuration.
- **Given** a Cinema belongs to a Location, **when** the Admin selects that Location for a showtime, **then** only its Cinemas are available.
- **Given** a Hall belongs to a Cinema, **when** the Admin selects that Cinema, **then** only its Halls are available.

VEN-02 Assign seat layout to hall

- **Given** predefined seat-layout templates exist, **when** the Admin configures a Hall, **then** the Admin can assign one of the predefined layouts.
- **Given** a Hall has no existing showtimes, **when** the Admin changes its layout, **then** the new layout becomes the Hall active layout.
- **Given** a Hall has been used by a showtime, **when** the Admin attempts to change its layout, **then** the system rejects the change.

SCH-01 Create and manage showtimes

- **Given** an active Movie, valid Location, Cinema, and Hall exist, **when** the Admin enters a valid showtime, **then** the system creates the showtime.
- **Given** a Hall has an existing showtime, **when** the Admin attempts to create an overlapping showtime in the same Hall, **then** the system rejects it.
- **Given** the movie duration is 120 minutes and the required buffer is 15 minutes, **when** the Admin attempts to schedule the next show before 135 minutes after the previous start time, **then** the system rejects it and explains the conflict.
- **Given** two different Halls in the same Cinema are available, **when** the Admin creates overlapping showtimes in those different Halls, **then** the system permits them.
- **Given** a showtime has no confirmed bookings, **when** the Admin removes it, **then** the system removes the showtime.
- **Given** a showtime has one or more confirmed bookings, **when** the Admin attempts to remove it, **then** the system rejects the operation.

BKG-01 View showtimes

- **Given** a Movie has configured showtimes, **when** a Customer selects a Location, **then** the system shows available Cinemas and showtimes for that Location.

- **Given** a showtime has all seats BOOKED, **when** the Customer views it, **then** the showtime remains visible, is visually disabled, and is labeled Sold Out.
- **Given** all seats are currently HELD but none are BOOKED, **when** the Customer views the showtime, **then** the system does not label it Sold Out and indicates that seats are temporarily unavailable.
- **Given** a showtime has at least one available seat, **when** the Customer views it, **then** the showtime can be selected.

BKG-02 View seat layout and status

- **Given** a Customer selects a valid showtime, **when** the seat-selection page opens, **then** the system displays the layout assigned to that Hall.
- **Given** a seat is AVAILABLE, **when** the seat map is displayed, **then** the seat is selectable.
- **Given** a seat is HELD by another customer or BOOKED, **when** the seat map is displayed, **then** the seat is visually unavailable and cannot be selected.
- **Given** a Customer has seats HELD for the current booking, **when** they return to the seat page, **then** the system identifies those seats as belonging to their active hold.

BKG-03 Hold selected seats

- **Given** E5 is AVAILABLE, **when** a Customer confirms selection of E5, **then** the system atomically changes E5 to HELD for that booking and starts a 10-minute hold.
- **Given** a seat is no longer AVAILABLE **when** the confirmation request is processed, **when** the Customer tries to confirm the seat, **then** the system rejects the request and informs the Customer that the seat is unavailable.
- **Given** a Customer's 10-minute hold expires without successful confirmed payment, **when** the hold expires, **then** all held seats return to AVAILABLE.
- **Given** payment FAILS while the hold is still active, **when** the failure is received, **then** the seats remain HELD until the original 10-minute timeout.
- **Given** the booking session is explicitly ended before payment succeeds, **when** the system terminates the booking session, **then** its held seats are released.

BKG-04 Concurrent seat reservation

- **Given** two Customers attempt to confirm the same AVAILABLE seat at approximately the same time, **when** both requests reach the booking system, **then** exactly one request successfully acquires the seat.
- **Given** one request successfully acquires the seat, **when** the second request is processed, **then** the second request is rejected.
- **Given** the second request is rejected, **when** the Customer receives the response, **then** the Customer is informed that the seat is no longer available and can return to seat selection.

PAY-01 Payment attempts

- **Given** a Booking has active held seats and no active payment attempt, **when** the Customer chooses Pay, **then** the system creates one active payment attempt.
- **Given** a Booking already has an active payment attempt, **when** the Customer clicks Pay again, **then** the system does not create a second active attempt.
- **Given** a payment attempt has failed and the booking hold is still active, **when** the Customer chooses to retry payment, **then** the system creates a new attempt and marks the previous attempt inactive.
- **Given** a new payment attempt is created, **when** it is sent to the provider, **then** it is associated with the same Booking and held seats.

PAY-02 Payment result and reconciliation

- **Given** the provider returns SUCCESS while the held seats are still valid, **when** the system verifies the result, **then** the Booking becomes CONFIRMED, the seats become BOOKED, and a ticket is generated.
- **Given** the provider returns FAILED, **when** the system receives the result, **then** the attempt becomes failed and the seats remain HELD until the original hold expires.
- **Given** the provider returns CANCELLED, **when** the system verifies the result, **then** the payment attempt stops and the seats are released.
- **Given** the provider returns UNKNOWN, **when** the result is received, **then** the system queries the provider for the authoritative transaction status before releasing the seats.
- **Given** the provider returns PENDING, **when** the payment remains pending, **then** the seats remain HELD until the defined payment/booking timeout.
- **Given** a payment remains PENDING until timeout, **when** the timeout is reached, **then** the system checks/cancels the provider-side transaction as appropriate and releases the held seats.
- **Given** a SUCCESS result arrives after the seat hold has expired, **when** the system verifies the payment, **then** if all requested seats are still AVAILABLE the Booking is confirmed; otherwise no ticket is issued and the system requests a refund/cancellation through the payment provider.

TCK-01 Ticket generation

- **Given** payment is successfully confirmed, **when** the Booking becomes CONFIRMED, **then** the system generates a ticket.
- **Given** a ticket exists, **when** the Customer opens it, **then** it displays the Movie, Cinema, Showtime, Seats, Customer name, Booking ID, QR code, and Price.
- **Given** a confirmed ticket exists, **when** the Customer attempts to cancel it, **then** the system rejects the cancellation because confirmed tickets are non-cancellable.

TCK-02 My Tickets and booking history

- **Given** a Customer has an upcoming confirmed ticket, **when** they open My Tickets, **then** the upcoming ticket remains visible regardless of **when** it was purchased.
- **Given** a Customer has a past confirmed ticket purchased within the previous 30 days, **when** they open booking history, **then** the ticket is displayed.
- **Given** a Customer has a past ticket older than 30 days, **when** they open the normal booking-history view, **then** that past ticket is not displayed.
- **Given** an archived movie is referenced by a ticket that remains in the Customer records, **when** the Customer opens that ticket, **then** the archived movie information remains accessible.

TCK-03 Ticket validation

- **Given** Cinema Staff is authenticated and scans a valid QR code, **when** the ticket belongs to the appropriate showtime and has not previously been used, **then** the system reports the ticket as valid.
- **Given** the QR code is invalid or does not correspond to a ticket, **when** Staff scans it, **then** the system rejects it.
- **Given** a ticket has already been validated, **when** Staff scans it again, **then** the system reports that it has already been used.
- **Given** a valid QR code belongs to a different showtime, **when** Staff scans it, **then** the system rejects it for the current showtime.

4. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Payment Security	Payment-provider communication shall use secure transport. Provider callbacks shall be authenticated using the provider signature/authentication mechanism, and the system shall verify payment reference, booking reference, expected amount, currency, and expected transaction state before accepting the callback.
NFR-02	Authentication Security	Passwords shall be stored using secure password hashing and never as plaintext. Protected endpoints shall require authenticated authorization.
NFR-03	Concurrent Seat Consistency	Seat reservation shall be atomic so that a seat cannot be successfully HELD or BOOKED by two customers simultaneously.
NFR-04	Seat Availability Response	Under the planned test load, seat-availability requests should complete within 2 seconds at the 95th percentile.
NFR-05	Concurrent Users	The MVP should support at least 50 concurrent customer booking sessions during testing and demonstration.
NFR-06	Payment Idempotency	Each payment attempt shall be idempotent: repeated requests for the same attempt must not create duplicate charges or duplicate ticket confirmation.
NFR-07	Historical Data Integrity	Historical bookings and tickets must retain their original movie, showtime, cinema, hall, and seat-layout relationships even after a movie is archived.
NFR-08	Seat Layout Integrity	A Hall seat layout shall remain immutable after the Hall has been used by a showtime.

5. MVP and Release Plan

Release	Scope
MVP	Identity, core catalog, venue configuration, scheduling, seat selection/holding, concurrency control, payment integration, ticket generation, and QR ticket validation.
Release 2	Password reset, movie archiving, and recent booking-history view improvements.

Core MVP journey: Register/Login → Browse Movie → View Details → Select Location/Cinema/Showtime → View Seat Layout → Hold Seats → Payment → Confirm Booking → Ticket/QR → Cinema Staff Validation.

6. Key Testing Scenarios

ID	Scenario	Expected behavior
TS-01	Successful booking	Customer completes registration/login, selects available seats, pays successfully, and receives a confirmed QR ticket.
TS-02	Sold-out showtime	All seats are BOOKED; the showtime remains visible but disabled and cannot be selected.
TS-03	Seat-hold expiration	Customer holds seats, does not complete payment for 10 minutes, and the seats return to AVAILABLE.
TS-04	Payment failure	Payment fails while the hold is active; seats remain HELD until the original 10-minute timeout.
TS-05	Duplicate payment click	Customer clicks Pay repeatedly; only one active payment attempt is created and duplicate ticket confirmation does not occur.
TS-06	Unknown payment result	Provider returns UNKNOWN; system queries authoritative provider status before deciding whether to release seats.
TS-07	Late payment success	SUCCESS arrives after hold expiry; system checks whether seats are still free and either confirms or initiates refund/cancellation handling.
TS-08	Concurrent seat request	Two customers confirm the same seat; exactly one request acquires the seat.
TS-09	Invalid schedule	Admin attempts overlapping or insufficiently spaced showtimes in the same hall; system rejects the schedule with a reason.
TS-10	Unauthorized access	Customer attempts to access Admin/Staff functions; system denies the request.
TS-11	QR validation	Cinema Staff scans a valid, invalid, already-used, and wrong-showtime QR code and receives the correct result.

7. Project Assumptions and Open Decisions

ID	Assumption / Decision
A-01	The project is a web application. Automated end-to-end testing is planned with TestCafe.
A-02	Locations are limited to Vietnam for the current project.
A-03	Cinema/Hall information is configured by the Admin based on information supplied by cinemas; the project does not require cinema-specific staff accounts for schedule management.
A-04	Admin and Cinema Staff accounts are provisioned/seeded rather than publicly registered.
A-05	Seat-layout templates are predefined. A Hall layout is locked after the Hall has been used by a showtime.
A-06	The showtime spacing buffer is currently specified as 15 minutes for the backlog examples; the team should confirm whether this is fixed for the final requirements.
A-07	Successful customer bookings are non-cancellable in the current scope.
A-08	Upcoming tickets remain visible beyond 30 days. The 30-day rule applies only to normal past-history visibility, not deletion of underlying records.
A-09	A showtime is considered Sold Out only when all seats are BOOKED; temporary HELD seats use a different unavailable state.